

Sleep, Sleep Disorders, and Mental Health: A Narrative Review

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Abstract

Evidence suggests that sleep is a vital component of physical and health well-being. However, while sleep problems are present in individuals with mental health problems such as depression, it has not been clear whether these conditions are independent or whether they might be causally related. Indeed, if sleep or sleep disorders predispose or modify onset and outcomes of mental health issues, treatment of these factors could be explored as new mental health prevention or treatment options. The aim of this review was to examine in detail the bidirectional relationship between sleep, sleep disorders, and mental and physical health and well-being. It has considered the evidence that sleep architecture disruption, occurring through both quantity, quality, and timing of sleep as well as through the presence of sleep disorders may both influence mental health and well-being as well as be disrupted by both physical and mental health conditions. Also, the review has explored the effects of sleep disruption on mental health and performance through fatigue, mood, and vigilance. The review has considered the bidirectionality between sleep, sleep disorders, and mental health to examine how these may lead to or exacerbate mental health disorders such as affective, anxiety, autism, depressive and schizophrenia disorders but also considers how these conditions can affect sleep. The review highlights that poor sleep or the presence of a sleep disorder can increase the risks from mental health conditions such as suicidality. Furthermore, mental health conditions such as anxiety and worry can cause racing or repetitive thoughts that can keep an individual awake, leading to shortened sleep. It is important that sleep and sleep disorders are considered potential modifiable factors that could improve mental health outcomes. The important interconnect between both physical and mental health and sleep, in patient evaluations, also needs to be considered as these may affect treatment pathways and patient outcomes. Further, more robust and perspective research is required to establish the triad relationship of physical, mental health, and sleep.

Keywords: Cardiovascular diseases, depression, mental health, seep, sleep disorders, sleep wake disorders

INTRODUCTION

Sleep is fundamentally important for good physical and mental health and well-being. Good sleep quality and quantity and absence of sleep disorders are also key components of emotional regulation,^[1] memory committal,^[2] physiological recovery,^[3] and relaxation.^[4] Sleep disturbances and, in particular, insomnia often co-occur with mental health disorders. Moreover, sleep disorders including insomnia, hypersomnia, and fatigue are part of the diagnostic criteria for depression. While it is not clear if these conditions are causally related or independent of one another, longitudinal studies have demonstrated that sleep disturbances often begin before the onset of depressive symptoms. Furthermore, sleep problems can affect the response to treatment in depression.^[5] However, there may be individual susceptibility to the adverse effects of disrupted sleep.

The aim of this review was to examine in detail the bidirectional association between sleep, sleep disorders, and both mental and physical health and well-being to determine if they are causally related [Appendix 1]. Thus, to then consider whether the treatment of sleep disorders might then be an important factor in the onset, progression or outcome of mental health conditions. Subsequently having key clinical implications for the effective treatment of these conditions.

SLEEP ARCHITECTURE DISRUPTION AND MENTAL HEALTH

Sleep is a complex arrangement of biological processes for neurological and physiological recovery. During sleep,

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the body goes through multiple sleep cycles. Split into two predominant sections, nonrapid eye movement sleep (NREM), or slow-wave sleep (SWS) of which has three stages (N1–N3) characterized by both light and deep sleep and one stage (N4) of rapid eye movement sleep (REM).^[6] REM sleep is a behavior state whereby most dreams occur; it is also a paradoxical state, with muscle paralysis and the suspension of consciousness despite cerebral cortex activation. REM sleep is, associated with memory and behavior development.^[7,8] REM cycles vary by length and by timing during the night, with the tendency to have more REM sleep later into the night.^[9]

The cycling of REM and NREM sleep is characterized through ultradian rhythms forming REM pressure.^[10,11] Although ultradian rhythms are themselves a rhythm based on sleep achievement, they are influenced by the drive to induce sleep through “Process C,” circadian rhythm, and the sleep-wake dependent homeostatic drive “Process S.”^[12]

Fragmentation of REM sleep has the potential to mark posttraumatic stress disorder (PTSD).^[13] In a large meta-analysis of polysomnography studies, the relationship between sleep architecture and mental health disorder was examined. The study indicated that sleep depth and REM pressure alterations were associated with affective, anxiety, autism, and schizophrenia disorders.^[14] Reduced SWS has also been associated with insomnia,^[15] mood, and anxiety disorders,^[16] generalized alterations of SWS are present in psychosis and neurodevelopmental conditions,^[17] and anxiety.^[14] Similarly, in adults with depression and schizophrenia, decreased slow-wave, or deep sleep, is associated with the severity of symptoms and hospitalization.^[18]

Sleep architecture is disrupted in depression, with disinhibition of REM sleep, and NREM sleep impairment with decreased amounts of N2 sleep. Some patients have less SWS, along with changes in sleep continuity, which might be linked to the hypothalamic–pituitary–adrenal (HPA) axis.^[19] A UK BioBank study of 91,105 participants demonstrated that the extent of disrupted circadian rhythmicity of rest-activity cycles was associated with an increased lifetime risk of depression and bipolar disorder, though the cause of this association is unknown.^[20]

An individual’s chronotype also determines their natural inclination to go to sleep. The preference can be split into going to bed early and getting up early (larks) or going to bed late and getting up late (owls). It is determined by several factors including genotype, age, and gender, although most individuals have an intermediate sleep schedule. It has been reported that individuals with the evening chronotype have an increased risk of depressive symptoms and depressive disorders.^[21]

In general, individuals have a circadian rhythm circle that regulates sleep and wakefulness which is >24 h. This means the internal biological clocks need to be constantly entrained to the external environment and 24-h day cycle. This regulation occurs via external cues (“Zeitgebers”) from sunlight and temperature.

The external cues influence circadian entrainment by the central clock or suprachiasmatic nucleus (SCN). The SCN is also influenced by the peripheral clocks which are in turn influenced by external cues of exercise or food and drink consumption and internal cues with circadian genes, such as *CLOCK*, *BMAL1*, and *PERIOD3*.^[22–24] The external and internal cues and central and peripheral clocks work in tandem to regulate physiological functions, wakefulness, and peripheral tissue activity.^[12,25,26]

CIRCADIAN SLEEP DISRUPTION AND MENTAL HEALTH

One of the main factors that entrain the clock is natural light. Disruption of this circadian clock can affect mood and emotional regulation^[27,28] and is also closely related to mental health disorders.^[28,29] Circadian disruption occurs when the environmental signals such as the phase of the light–dark cycle are misaligned with the internal body clocks. This may occur because of shift work or travelling through time zones or because of having different sleep times on work and social days (social jet lag)^[29] or through seasonal disruption and clock time changes.

Shift work

Shift work is often associated with working a night shift, which leads to a later onset of sleep than normal circadian rhythms working against the drive for sleep. A recent meta-analysis demonstrated that nurses working night shifts had an increased likelihood of depression.^[30] In a separate study, shift workers were shown to have a high prevalence of anxiety and depression.^[31] Parasomnias are also reported to be more prevalent in rotating shift and night shift workers compared to daytime shift workers.^[32] Poorer moods and well-being are also reported in shift workers compared to non-shift workers.^[33] In a UK BioBank study, shift working was shown to increase the risk of depression and anxiety. However, among the shift workers, there was no significant difference due to night shifts or nonnight shifts.^[34]

Travel and social jet lag

Airline travel can lead to sleep disruption and jet lag. Often traveling quickly across time zones causes jetlag as individuals are eating and sleeping out of synchrony with their normal circadian rhythm.^[35,36] Early evidence from a study conducted in Israel demonstrated that psychiatry episodes increased with jet lag; foreign patients were more likely to have psychiatry episodes with flights over 7 h or when more than three time zones were crossed.^[37] Flight attendants in the USA have been found to have greater levels of sleep disorder and depression compared to the general population.^[38] More recent research demonstrated that international air crew had increased levels of depression symptoms compared to domestic air crew. Jet lag is implicated in mood and performance among athletes, who need to travel.^[39,40]

Social jet lag can occur without travel and is often used to describe the feeling that occurs when a person, who has slept in longer at the weekend, has to get up early on a weekday for work.^[41] In adolescents, the misalignment between sleep timing

on the weekend and school week, resulting in social jet lag, was independently associated with anxiety symptoms.^[42] Office workers demonstrating social jet lag on a weekend were more likely to report job stress.^[43] In South Korea, social jet lag was associated with increased risk of depressive symptoms, even more when the social jet lag was over 2 h.^[44] Further research suggests that greater levels of social jet lag are associated with increased symptoms of depression across 20 days.^[45] Japanese research has demonstrated that social jet lag and sleep debt worsen depressive symptoms with a “U-shaped” distribution of sleep duration and depressive score.^[46]

Seasonal disruption and clock changes

During the year, there are seasonal variations in the amount of sunlight and the weather. Light plays in circadian rhythms, so it is not surprising that there are adjustments in sleep quality and quantity throughout the year. The dark hours in the Arctic Circle have been associated with greater delayed sleep and impaired mood.^[47] Further, the amount of light, which varies over the seasons, impacts mood and sleep through the SCN.^[48] Between March and November in the US and UK, the clocks move forward ahead of the time zone. This is known as daylight saving time in the US and British summer time in the UK. The acute variation in time and the changing of the clocks is thought to be associated with changes in sleep, mood disturbance, and suicide; acute sleep deprivation in March due to the clocks moving forward is likely to affect the frontal lobe and the ability to make judgments.^[49] Over 70 countries have similar biannual changes in time, evidence suggests that it impacts sleep and mood especially when clocks move more than an hour forward.^[50] Further evidence suggests that moving the clocks forward worsens sleep quality, decreases sleep, and can increase depression.^[51] Additionally, in a 7-year study, permanent daylight saving and clocks being ahead of time for Russian school children increased seasonal affective disorder (SAD).^[52]

Seasonal affective disorder

SAD is characterized as a mood disorder with an onset consistent with the autumn and winter months which resolved in the spring and is related to sleep timing changes and hypersomnolence.^[53] Early research demonstrated that SAD is defined by the hypersomnia in the winter months, with decreased delta wave sleep and increased REM density.^[54] Evidence also suggested that a lower core body temperature and lower melatonin secretions are a key in the circadian disruption in SAD.^[55] A recent meta-analysis has demonstrated that there is altered REM sleep amount and latency but no altered SWS in SAD.^[56]

SLEEP LOSS, QUALITY AND QUANTITY, AND MENTAL HEALTH

An individual's requirement for sleep may vary, but in general, it is expected that an adult requires around 7 h of sleep per night.^[57] A bad night's sleep due to insufficient, poor quality, or mis-timed sleep leads to fatigue, depressed mood, impaired

performance, and vigilance.^[58–60] In total sleep deprivation laboratory studies, individuals may, at first, demonstrate sleepiness, anxiety, and decreased emotional capacity, but after 24–48 h, this may lead to temporal disorientation and hallucinations, and after 72 h, it may result in impaired sensory responses and acute psychosis.^[61]

Several epidemiological studies have looked at the effect of sleep on mental health and it has been suggested that sleep deprivation is associated with major depressive disorder^[62] and that short sleep and insomnia may lead to a high risk of depression.^[61] In older adults, a meta-analysis has demonstrated that sleep quality is associated with probable depressive symptomatology.^[63] In participants ($n = 84,404$) from the UK Biobank, it was shown that a “U-shaped” relationship was found between sleep duration and diseases including diabetes, hypertension, heart disease, and depression; short and long sleep durations, along with fragmented sleep, were associated with increased odds of disease.^[64] In the English Longitudinal Study of Aging, short sleep (<5 h) over an 8-year follow-up increased the risk of depression.^[65] A recent meta-analysis suggested that sleeping less or more than 7 h compared to sleeping compared to 7 h of sleep was associated with higher levels of depression.^[66] In the USA, a cross-sectional study reports a U-shaped distribution of sleep and risk of depression with both short sleep and long sleep associated with depression.^[67] Also, analysis of UK BioBank data shows a U-shaped association between sleep length and mental health, with those sleeping six to 8 h a night having fewer mental health conditions compared to those sleeping less than six hours and more than eight hours.^[64]

Among a cardiovascular health study, sleep disturbances relating to difficulty falling asleep, frequent awakenings, waking up too early, and daytime sleepiness were associated with depression.^[68] The glymphatic system is a key waste-removal system that is effective and most prominent in slow sleep waves; both mood disorders and sleep disruption show alterations in the glymphatic system. In major depression disorder, glymphatics disruption impairs the removal of reactive oxygen species.^[69] This signals a potential mechanism for sleep disruption and mental health disorders. The glymphatic dysfunction is associated with neurological disorders.^[70] Further research shows that sleep is associated with oscillations that influence the cerebrospinal fluid (CSF) flow.^[71]

In a study looking at sleep disturbances and the AtRisk Mental State (ARMS) in youth, it was shown that ARMS individuals report impaired sleep quality and reduced sleep quantity compared to healthy controls. Furthermore, sleep disturbances were associated with attenuated psychotic symptoms and functional outcomes cross-sectionally and longitudinally.^[72] Beyond clinical and subclinical mental health conditions, sleep length has been associated with changes in mood. In adolescents, deeper sleep and higher sleep efficiency are associated with greater mood.^[73] Moreover, a meta-analysis in adolescents demonstrated that sleep duration has the

largest relationship with positive moods, when being linked to anger, depressive, negative affect, and anxiety.^[74] Deeper and higher sleep efficiency are associated with happier moods in adolescents.^[73]

An experimental study of male teenagers found that restricting sleep to 5 h decreases happiness and increases anger, confusion, and depression. Importantly it took more than 2 days for their mood to recover to their baselines. In the same study, teenagers sleeping for 10 h saw an increase in happiness.^[75] A study of sleep duration across the lifespan demonstrated that sleep loss through deprivation or restriction increased negative mood and emotions.^[76] Sleep length might interact with the underlying sleep architecture which could influence mood.

Sleep length and mental health is influenced further by social factors and was particularly influenced by the COVID-19 pandemic. In a US study, even after adjustment for covariates (age, sex, race, education, income, and body mass index), in the 1st year of the pandemic, it was found that higher socioeconomic deprivation was associated with shorter sleep and poorer mental health. Individuals in areas of the highest socioeconomic deprivation were 60% more likely to sleep 6 h or less and 50% more likely to have more than 14 days of poor mental health.^[77]

Sleep loss and sociability

Sleep loss is associated with loneliness, social isolation, and withdrawal which could signal that sleep influences the regulation of interpersonal interactions.^[78] In close social relationships, sleep quality can predicate the nature of the tie between the individuals and mental health conditions can influence this relationship. Poor sleep which is associated with aversive ties and good sleep quality was associated with supportive ties. Depression increases poor sleep which increases the aversive friendship ties.^[79] Sleep difficulties and sleep loss resulting in social isolation are an important consideration given loneliness is associated with psychiatric disorders.^[80] Recent research has also demonstrated that sleep duration can decrease socialization and social activity can predict sleep length and sleepiness. Social activity in the middle of the day has the greatest increase in sleepiness.^[81]

Sleep loss and emotional regulation

Bidirectional relationships in processing stress, emotions, and sleep exist. Sleep gives the capacity to process stress and emotions. Whilst decreased sleep quality and quantity decreases the capacity to manage stress and emotions. Poor sleep has been directly correlated with stress, anger, hostility, and verbal aggression.^[82] The emotional regulation with sleep also extends into psychiatric disorders with emotional functional deficits with insufficient sleep.^[83] Early research pointed toward sleep loss disconnecting the amygdala from the prefrontal cortex.^[84] The association of sleep and emotional function was further synthesized. REM sleep allows for the consolidation of affective information in subconscious memory. Therefore, REM sleep is important for homeostasis of affective brain function.^[85]

SLEEP DISORDERS AND MENTAL HEALTH

Sleep disorders have an impact on mental health, but likewise mental health disorders can influence sleep (see next section). In army veterans, those with serious mental health disorders are more likely to have a sleep disorder.^[86] Nightmares and dreams have also been linked to a relationship of mental illness.^[87] Moreover, sleep disturbances and sleep disorders potentially can lead to further neurodegenerative disease.^[88]

Insomnia

Insomnia is a sleep disorder characterized by difficulty falling asleep, staying asleep, or both. As well as, being a sign of mental health conditions, it can be a predictor for depression.^[89] This reinforces the importance of emotional regulation during sleep. Insomnia has also been associated with increases in cardiovascular disease (CVD) and CVD mortality.^[90] In a 10-year cohort study, insomnia symptoms increased the risk of ischemic heart disease and was associated with those with hypertension.^[91] A recent review of meta-analyses of prospective studies investigating the association between insomnia and multiple health outcomes showed that insomnia was mostly related to cardiovascular outcomes (atrial fibrillation, CVD, coronary heart disease, myocardial infarction, and stroke) and mental health disorders (anxiety, depression, suicidal ideation, attempts, and death). Insomnia was also associated with an enhanced Alzheimer's disease risk.^[92] Insomnia has a bidirectional relationship with emotional dysregulation. It is thought insomnia disinhibits the limbic system, which tips the balance towards mood and psychiatric disorders. In turn, mental health conditions influence the prevalence of insomnia. Importantly, digital technology also increases insomnia.^[93]

A review of early literature has suggested that sleep disturbances relate to suicide risk in mental health disorders. In some cases, insomnia and nightmares showed a unique risk of suicidality.^[94] More recent meta-analyses demonstrate that sleep disturbances and insomnia increase suicide ideation.^[95-97]

In depression, insomnia is an indicator of suicidality.^[98] Also delayed sleep timing increases suicidal ideation in depression with insomnia.^[99] In schizophrenia, suicide attempts increased with insomnia.^[100] Early evidence has demonstrated that nightmares have a five times greater risk for suicidality.^[101] More recent meta-analysis of suicide attempt was 81% higher in those experiencing nightmares.^[96,97,102] In adolescents, meta-analyses also demonstrate that suicide ideation and attempts were increased in sleep disturbances, and depression did not moderate this relationship.^[103]

Obstructive sleep apnea

Obstructive sleep apnoea (OSA) is characterized by complete or partial obstruction of the airway resulting in intermittent hypoxia. OSA has been related to mental health conditions and mood.^[104,105] In their review, Gupta and Simpson^[104] suggested psychotic conditions develop from sleep fragmentation caused by OSA, as it induces inflammation and neurotransmitter imbalance. This is thought to dysregulate the sympathetic

nervous system and HPA axis. Garbarino and Magnacvita^[106] found that in the workplace, there was an association with metabolic syndrome, OSA, and psychological disorders. A recent systematic review of OSA patients and health-related quality of life found that those who have untreated OSA have lower emotional adjustments and poorer psychologic adjustment to illness.^[107] A study of 8,030 individuals in Korea also demonstrated that OSA was associated with a higher risk of both depression and stress.^[108]

OSA is also associated with suicide risk and self-harm. Within a US survey, sleep apnea was associated with self-reports of suicidal ideation, planning, and attempts.^[109] In Denmark, the sleep apnea increased the risk of suicide and self-harm.^[110] In Taiwan, similarly, sleep apnea was associated with an increased risk of suicidal attempt.^[111] Furthermore, in Taiwan, adolescents suspected of having pediatric OSA had an increased risk of suicidal ideation.^[112] Patients with both major depressive disorder and OSA in a longitudinal study in the US demonstrated that there is an increased risk of suicidal ideation and attempt; in both groups of OSA with or without depression, suicidal ideation was above 40%.^[113] OSA was also found in 25% of 125 patients who had major depressive disorder and suicidal ideation.^[114] Further in comorbidity PTSD and OSA, the severity of OSA was associated with suicidal ideation.^[115] Social support is key in mental health and suicidality, but in 149 OSA patients, where 20.1% had suicidal ideation, social support did not make a difference to their suicidality.^[116]

The most common form of treatment for OSA is with continuous positive airway pressure (CPAP). In their meta-analysis of nine CPAP studies, Yang *et al.*^[117] demonstrated that 4 h of CPAP per night improved mental health symptoms. Evidence in mice studies suggests CPAP treatment restores the glymphatic system and CSF flow.^[118] Human CPAP studies are required to confirm such associations and whether this is linked to the improvements in mental health.

Untreated OSA can affect not only the sleep and daytime functioning of an individual with the condition but also that of a bed partner.^[119] Bed partners of untreated OSA have increased depressive scores.^[120] It has been shown that when the condition is treated with CPAP, bed partner anxiety and depression decreases.^[121]

Comorbid insomnia and sleep apnoea

Insomnia and sleep apnea can co-occur as Co-Morbid Insomnia and Sleep Apnoea (COMISA). The prevalence of insomnia in OSA ranges between 6% and 84% and OSA in insomnia ranges between 7% and 69%.^[122,123] In a recent Australian study, mental health conditions were significantly higher in COMISA patients than controls.^[124] In men, COMISA has a greater prevalence and severity of depression compared to OSA and insomnia alone.^[125] Research across both genders shows increased depression with COMISA.^[126] In a study of 117 subjects with untreated OSA, it was found that the overall prevalence of suicidal ideation was 20.5%. For subjects who reported insomnia, the severity of insomnia was positively

correlated with depressive mood and suicidal ideation. However, after adjusting for depressive symptom severity insomnia and suicidality were insignificant. This study suggests the necessity of an integrated treatment approach to suicidal ideation in patients with COMISA.^[127]

Parasomnias

The abnormal movement, nightmares disorders, REM sleep behavior disorder, and sleep eating are collective parasomnias. Except for REM sleep behavior disorder which is linked to Parkinson's disease,^[128] the remaining parasomnias are linked with anxiety, stress, and emotions.^[129-131] Parasomnias are split into REM and NREM parasomnia.^[132]

Sleep paralysis a form of REM parasomnia is a temporary inability to control the body. It occurs in the hypnopompic (before waking up) or hypnagogic (before falling asleep) stage of sleep. Sleep paralysis has been associated with anxiety and PTSD.^[133] A systematic review of sleep paralysis has associated the experience with stressful life events, anxiety, and depression symptoms.^[130] Further discussion suggests sleep paralysis is related to poor sleep and nightmares.^[134]

Nightmare disorders are also associated with REM sleep.^[132] In a systematic review, nightmares were increased in mood disorders and psychosis disorders.^[135] Further evidence suggests that nightmares are associated with PTSD.^[136] A recent follow-up study of 40,902 participants found that having nightmares was associated with higher incidence rates of depression.^[137]

Sleep terrors, or night terrors, are associated with NREM sleep. It involves the vocalization of panic and terror and movements while not aroused.^[138] Sleep terrors are associated with childhood and PTSD. In children, night terrors, alongside nightmares, can increase the odds of experiencing later adolescent psychosis.^[139] More recent evidence in children has shown that sleep terrors were associated with more reactive behavior, anxiety, and depression symptoms.^[140] Beyond childhood, sleep terrors are frequently associated with PTSD, and a sign of PTSD that requires attention.^[141,142]

Narcolepsy

Narcolepsy is a rare neurological disorder interrupting the normal sleep circles causing periods of involuntary sleep, reduced REM sleep, and sleep paralysis. It is also associated with catalepsy, the involuntary sleep induced by emotions. In a research from Spain, patients with narcolepsy had triple the rate of anxiety and depression compared to the general population.^[143] In 9,312 patients in the United States, compared to controls, patients with narcolepsy had higher rates of mood disorders, depression, and anxiety.^[144] In a recent systematic review and meta-analysis, patients with narcolepsy have a significantly lower level of mental health and emotional quality of life score than the general population, with the review also demonstrating that narcolepsy was associated with lower mental health quality life scores compared to epilepsy, multiple sclerosis, diabetes, and hypertension.^[145]

Some evidence in case reports also describes narcolepsy being present in diagnosed psychiatric disorders, increasing the need to consider sleep assessment especially in nonimproving psychiatric disorders with excessive daytime sleepiness.^[146] In cataplexy, the emotional regulation is believed to be impaired for narcolepsy patients.^[147]

Restless leg syndrome

A neurological disorder that frequently impacts sleep, RLS is the urge to move the legs and is accompanied with an unpleasant crawling or creeping sensation.^[148] RLS disproportionately affects women.^[149] Depression and anxiety have been found to be higher in RLS than healthy controls.^[150] In the Wisconsin Sleep Cohort, daily symptoms were associated with elevated depression and anxiety.^[151] In a Swedish cohort, RLS has been associated with a greater risk of depressive symptoms among 2,608 men.^[152] In a recent Danish study, RLS is associated with an increased risk of depression and decreased health-related quality of life, and poor sleep is also detrimental for RLS, suggesting a possible triadic link.^[153] Kallweit *et al.*^[154] also report a worse quality of life for psychiatric conditions when they had RLS. An increased risk of suicide and self-harm has been shown in RLS patients within the United States.^[155] Moreover, in a study of 549 untreated patients with RLS, it was found that the frequency of depressive symptoms (32.5%) and suicidal thoughts (28%) was 10-fold and 3-fold higher, respectively, in patients with RLS than controls. Suicide risk was associated with depression, impulsiveness, and RLS severity. Whilst RLS treatment improved depressive symptoms it did not improve the suicidal thoughts. There is an importance in detecting these symptoms in young women to improve RLS management.^[156]

INTERRELATIONSHIP BETWEEN SLEEP, MENTAL HEALTH, AND PHYSICAL HEALTH CONDITIONS

Given the bidirectional associations between sleep and mental health, we will now also consider their relationship with physical health [Figure 1].

Cardiovascular conditions

The relationship between sleep and CVD risk is well established, with shortened sleep and poor-quality sleep.^[157] Further, there is evidence to suggest that anxiety, depression, PTSD, and schizophrenia increase the risk of coronary heart disease^[158,159] and that those who have CVD may have increased risk of poor sleep and mental health. In Canada, a prospective study in 33,445 individuals demonstrated that a sleep disorder and depressive symptoms increased the risk of developing heart disease.^[160] In a cross-sectional study, poor sleep has been shown to be associated with depression and anxiety among individuals with CVD, but further longitudinal studies are required to demonstrate any cause–effect relationships.^[161] In a recent Chinese study of 348, sleep disturbances with anxiety and depression were greater for stable angina than those with acute coronary syndrome.^[162] With the interaction of sleep, mental and CVD there is a need

to consider the physical health, sleep, and mental health of all patients. With especially attention those who are less likely to look after their well-being.

Neurological conditions

Ischemic and hemorrhagic strokes are a leading cause of death and disability. Patients who have had a stroke can report frequent rates of sleep disordered breathing, insomnia, hypersomnia, and poor sleep quality.^[163,164] Sleep disorders like sleep apnea and RLS have an adverse effect on functionality after a stroke.^[165] Furthermore, poor sleep has been shown to have an adverse impact on their rehabilitation.^[166,167] Following a stroke, depression and anxiety are also reported. Together both have a relationship after a stroke. In 1,619 Chinese stroke patients, persistent sleep disturbance is associated with an increased risk of both depression and anxiety.^[168] In 100 community patients, 4 months after a stroke, it has been reported that mood impacts following the stroke were related to sleep impairment.^[169] Evidence from 1,369 patients suggests that sleep duration <6 h increases the risk of depression after a stroke.^[170] In a study of both stroke survivors and their caregivers, poor sleep quality and depressive symptoms were correlated in both the survivors and caregivers.^[171]

Concussion

Concussion, an acute brain injury caused by a directed or indirect exposure of forces to the head, can both cause sleep and mental health disturbances.^[172] Theadom *et al.*^[173] found that when there was poor quality sleep, anxiety and depression were significantly higher compared to those recovering from a concussion without poor sleep. Previous sleep disorders diagnosed within concussion athletes increased the levels of anxiety and depression.^[174] Some evidence has also suggested that the co-occurrence of mental health symptoms and daytime sleepiness increases symptoms in postconcussion syndrome.^[175]

Cerebral palsy

In neurological disabilities, sleep and mental health also overlap. In cerebral palsy, sleep quantity and quality is also related to motor function and visual impairment.^[176] Anxiety and depression are higher in children with cerebral palsy and evidence suggests that shortened sleep duration increases the risk of anxiety and depression. The risk of anxiety and depression further increased with decreased physical activity.^[177] Lang *et al.*^[176] reported that sleep problems in children with cerebral palsy were related to poorer caregiver sleep quality. The poorer quality of sleep for caregivers in turn was related to poorer psychological health and well-being for them. Increased nighttime disturbances for caregivers of children with cerebral palsy have also been associated with increasing maternal depression.^[178]

Dementia

Within the context of dementia, sleep and mental health have similar relationships with sleep and mood. Treatments centered around sleep can improve mood. There is evidence of a relationship between sleep, mood disorders, and dementia;

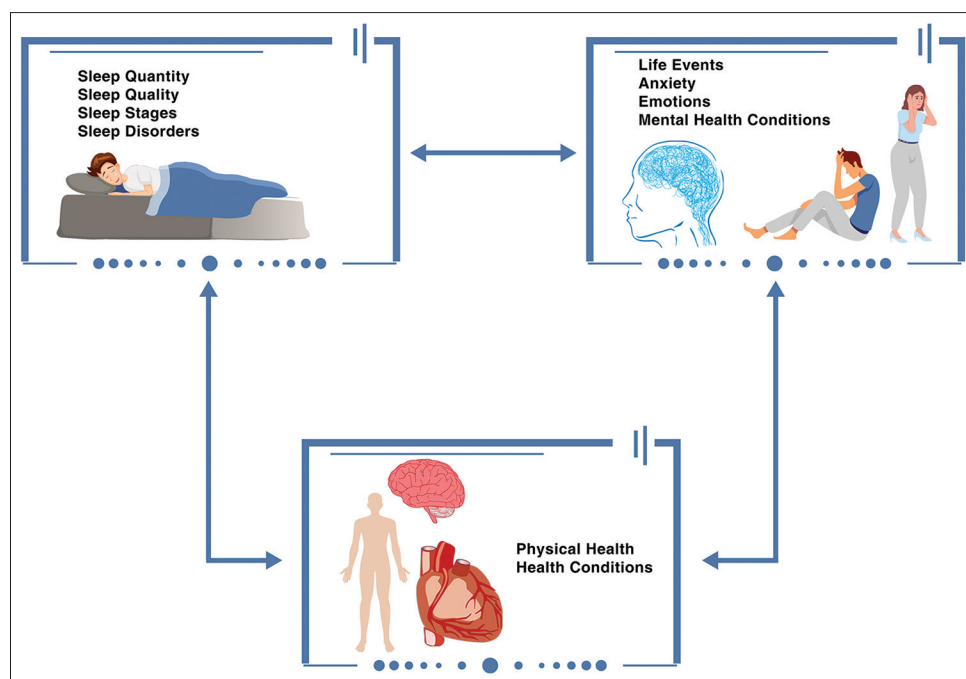


Figure 1: A representation of the different interactions between sleep and mental health and physical health conditions

in the UK BioBank those with sleep <6 h and had a low mood index had a higher risk of dementia.^[179] Sleep and mood for those with dementia is also important given the globalized impairment experienced within dementia. Light therapy for Alzheimer's disease and related diseases improves both mood and sleep quality.^[180] Similar results were shown in a systematic review of earlier studies around light treatment improving sleep duration, mood, and cognition in Alzheimer's.^[181] Although there is limited evidence on sleep and mood alone in dementia patients, light therapy resulting in improvements in sleep, mood, and circadian rhythms,^[182,183] highlights the involvement of sleep in dementia symptomatology.

Epilepsy

Encompassing various forms of complex neurological disorders, epilepsy broadly represents the seizure activity induced by the dysfunction in brain waves. Seizure activity frequently occurs overnight, disrupting sleep. Meta-analysis of children with epilepsy demonstrated that behavior difficulties can be related to sleep disruption and excessive daytime sleepiness, which might also be caused by coexisting sleep disorders.^[184] A recent meta-analysis of sleep disruption in adolescents and children found that sleep disturbance is common and so are a vast array of sleep disorder and highlights the potential impact on quality of life.^[185] Adults in Ethiopia with epilepsy who have anxiety and experience stressful life events were more likely to have adverse sleep quality which can cause seizure activity.^[186] Furthermore, sleep disturbance in those with epilepsy decreases their mental health and emotional well-being.^[187]

Physical pain

During sleep, the brain undertakes numerous neurological activities that are vital in mood, mental health, and neural

sensitivity. This gives a bidirectional relationship between pain and sleep impacting mental health. Those defining themselves as poor sleepers have been shown to have increased pain sensitivity and attention to pain. The increase in pain in poor sleepers increases depressive symptoms.^[188] Those with chronic physical pain have increased sleep complaints and depressive symptoms. This suggests there is a link in the regulation of emotions through sleep is important physical pain and internalizing symptoms.^[189] Other research also suggests that nighttime sleep is associated with the next day's mood and pain.^[190] Chronic pain and mood complaints are also associated with insomnia and sleep complaints.^[191] Improved sleep quality can pain and depression symptoms, which reinforces that sleep plays a key role in mental health and physical wellbeing.^[192] The role sleep plays in mental health continues in perioperative health. Recovery from operations increases with sleep disorders and higher levels of mental disorders.^[193] In a recent study of bariatric surgery, sleep disturbance among patients was associated with self-harm.^[194] Nonpharmacological sleep treatments in chronic pain patients are associated with large improvements in sleep quality, small reductions in pain, and moderate improvement in fatigue. This suggests nonpharmacological sleep interventions may represent a fruitful avenue for optimizing treatment outcomes in patients with chronic pain.^[195]

Women's health

Across the menstrual cycle, women can experience changes in sleep. Evidence suggests changes in sleep quality in the luteal phase, with increased sleep fragmentation.^[196] In a Korean population based-study, women who sleep <5 h a night had increased levels of depressive mood with irregular menstruation.^[197] Mood is often reported as fluctuating

during menstruation. A recent study of 72 women found good-quality sleep related to more positive moods during menstruation.^[198] Those experiencing premenstruation syndrome and premenstruation mood disorder can experience symptoms up to several weeks before menstruation and can experience insomnia.^[199] Further insomnia is frequent in premenstruation dystrophic disorder, a condition which effects mood and has been shown to have the greatest effect in the late-luteal phase.^[200]

Sleep requirements for women dramatically change during the menopause as opposed to the continuous decline in sleep requirements in males.^[19] In and around the menopause sleep disorders increase especially insomnia and sleep disordered breathing.^[201-203] The menopause is often related to mood changes and increased levels of anxiety and depression. In Italy, menopausal women with insomnia were associated with greater prevalence of mood disorders.^[204] In a recent Taiwanese study, depressive symptoms were associated with longer sleep-onset latency and lower sleep efficiency, mimicking the potential insomnia.^[205] In Brazil, more severe menopause symptoms were associated with poorer quality sleep and a higher prevalence of mental health disorders.^[206]

TREATMENT OF SLEEP AND SLEEP DISORDERS IN MENTAL HEALTH CONDITIONS

A recent systematic review of trials investigating the use of cognitive behavioral therapy for insomnia (CBT-I) for the treatment of depression has demonstrated that CBT-I presents a promising treatment option. It suggested that while this appeared to be the most effective when delivered via in-person sessions. More research was needed in this area.^[207] Furthermore, in another systematic review and meta-analysis, it has been suggested that CBT-I as well as being an effective treatment in patients with both insomnia and a single mental health condition (such as depression, PTSD, and alcohol dependency, is effective with mixed mental health disorders.^[208] It has been suggested that CBT-I should be considered as a first-line treatment in individuals with conditions such as depression or PTSD, especially given the side effects of medication. Research has suggested that treating OSA improves mental health symptoms among those with depression^[117] so should be screened for. However, more research is required to consider OSA diagnosis and treatment in individuals with other mental health condition.

CONCLUSION AND RECOMMENDATIONS

While there are emerging data to suggest that the treatment of sleep disorders might have a benefit on mental health disorders, more research is needed in this area. Given the bidirectional associations between sleep, sleep disorders, mental, and physical health and that there are many comorbid associations, prospective research is needed. Research should prospectively concentrate on the triadic relationship between sleep, sleep disorders, mental health, and physical health. There must be an assessment and treatment of sleep and sleep disorders in

mental health conditions and physical health conditions so that appropriate treatment regimens are developed. The research should attend to how that triadic relationship also impacts health outcomes and recovery in the long-term for patients.

Author contributions

Both authors designed and defined the criteria for the review. Both contributed to the selection of papers for inclusion in the manuscript. Both drafted, reviewed, and edited the document before submission. The final manuscript was reviewed, edited, and approved by both authors.

Ethical statement

Ethical statement is not applicable for this article.

Data availability statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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Conflicts of interest

Mr. Nathan E. Howarth is an executive member of the British Sleep Society. Prof. Michelle A. Miller receives royalties from Oxford University Press for two textbooks. She is Chief Investigator on the NIHR FOUND Trial (NIHR203393). She is also an executive member of the British Sleep Society. There are no conflicts of interest.

REFERENCES

1. Vandekerckhove M, Wang YL. Emotion, emotion regulation and sleep: An intimate relationship. *AIMS Neurosci* 2018;5:1-17.
2. Conte F, Carobbi G, Errico BM, Ficca G. The effects of pre-sleep learning on sleep continuity, stability, and organization in elderly individuals. *Front Neurol* 2012;3:109.
3. Vyazovskiy VV. Sleep, recovery, and metaregulation: Explaining the benefits of sleep. *Nat Sci Sleep* 2015;7:171-84.
4. Zielinski MR, McKenna JT, McCarley RW. Functions and mechanisms of sleep. *AIMS Neurosci* 2016;3:67-104.
5. Weich S. The epidemiology sleep and depression. In: Cappuccio FP, Miller MA, Lockley SW, editors. *Sleep, Health and Society From Aetiology to Public Health*. Oxford, UK: Oxford University Press; 2010. p. 178-90.
6. McCarley RW. Neurobiology of REM and NREM sleep. *Sleep Med* 2007;8:302-30.
7. Blumberg MS, Lesku JA, Libourel PA, Schmidt MH, Rattenborg NC. What is REM sleep? *Curr Biol* 2020;30:R38-49.
8. Mukai Y, Yamanaka A. Functional roles of REM sleep. *Neurosci Res* 2023;189:44-53.
9. Della Monica C, Johnsen S, Atzori G, Groeger JA, Dijk DJ. Rapid eye movement sleep, sleep continuity and slow wave sleep as predictors of cognition, mood, and subjective sleep quality in healthy men and women, aged 20-84 years. *Front Psychiatry* 2018;9:255.
10. Kishi A, Yasuda H, Matsumoto T, Inami Y, Horiguchi J, Tamaki M, *et al*. NREM sleep stage transitions control ultradian REM sleep rhythm. *Sleep* 2011;34:1423-32.
11. Le Bon O. Relationships between REM and NREM in the NREM-REM sleep cycle: A review on competing concepts. *Sleep Med* 2020;70:6-16.
12. Saper CB, Cano G, Scammell TE. Homeostatic, circadian, and emotional regulation of sleep. *J Comp Neurol* 2005;493:92-8.
13. Mellman TA, Bustamante V, Fins AI, Pigeon WR, Nolan B. REM sleep and the early development of posttraumatic stress disorder. *Am J Psychiatry* 2002;159:1696-701.

14. Baglioni C, Nanovska S, Regen W, Spiegelhalter K, Feige B, Nissen C, *et al.* Sleep and mental disorders: A meta-analysis of polysomnographic research. *Psychol Bull* 2016;142:969-90.
15. Hogan SE, Delgado GM, Hall MH, Nimgaonkar VL, Germain A, Buysse DJ, *et al.* Slow-oscillation activity is reduced and high frequency activity is elevated in older adults with insomnia. *J Clin Sleep Med* 2020;16:1445-54.
16. Palagini L, Baglioni C, Ciapparelli A, Gemignani A, Riemann D. REM sleep dysregulation in depression: State of the art. *Sleep Med Rev* 2013;17:377-90.
17. Aquino G, Alfi G, Riemann D, Laurino M, Menicucci D, Piarulli A, *et al.* Sleep is essential for mental health: Potential role of slow oscillations. *Curr Sleep Med Rep* 2024;10:1-10.
18. Drews HJ, Wiesner CD, Bethke-Jaenicke C, Weinhold SL, Baier PC, Göder R. Slow-wave sleep predicts long-term social functioning in severe mental illness. *PLoS One* 2018;13:e0202198.
19. Steiger A, Pawlowski M. Depression and sleep. *Int J Mol Sci* 2019;20:607.
20. Lyall LM, Wyse CA, Graham N, Ferguson A, Lyall DM, Cullen B, *et al.* Association of disrupted circadian rhythmicity with mood disorders, subjective wellbeing, and cognitive function: A cross-sectional study of 91 105 participants from the UK Biobank. *Lancet Psychiatry* 2018;5:507-14.
21. Haraden DA, Mullin BC, Hankin BL. The relationship between depression and chronotype: A longitudinal assessment during childhood and adolescence. *Depress Anxiety* 2017;34:967-76.
22. Dardente H, Cermakian N. Molecular circadian rhythms in central and peripheral clocks in mammals. *Chronobiol Int* 2007;24:195-213.
23. Albrecht U. Timing to perfection: The biology of central and peripheral circadian clocks. *Neuron* 2012;74:246-60.
24. Mohawk JA, Green CB, Takahashi JS. Central and peripheral circadian clocks in mammals. *Annu Rev Neurosci* 2012;35:445-62.
25. Abbott SM, Malkani RG, Zee PC. Circadian disruption and human health: A bidirectional relationship. *Eur J Neurosci* 2020;51:567-83.
26. Lane JM, Qian J, Mignot E, Redline S, Scheer FA, Saxena R. Genetics of circadian rhythms and sleep in human health and disease. *Nat Rev Genet* 2023;24:4-20.
27. Boivin DB, Czeisler CA, Dijk DJ, Duffy JF, Folkard S, Minors DS, *et al.* Complex interaction of the sleep-wake cycle and circadian phase modulates mood in healthy subjects. *Arch Gen Psychiatry* 1997;54:145-52.
28. Meyer N, Lok R, Schmidt C, Kyle SD, McClung CA, Cajochen C, *et al.* The sleep-circadian interface: A window into mental disorders. *Proc Natl Acad Sci U S A* 2024;121:e2214756121.
29. Walker WH 2nd, Walton JC, DeVries AC, Nelson RJ. Circadian rhythm disruption and mental health. *Transl Psychiatry* 2020;10:28.
30. Okechukwu CE, Colaprico C, Di Mario S, Oko-Obob AG, Shaholli D, Manai MV, *et al.* The relationship between working night shifts and depression among nurses: A systematic review and meta-analysis. *Healthcare (Basel)* 2023;11:937.
31. Li Y, Wang Y, Lv X, Li R, Guan X, Li L, *et al.* Effects of factors related to shift work on depression and anxiety in nurses. *Front Public Health* 2022;10:926988.
32. Kurt Gök D, Ünal İ, Aslan-Kara K. Evaluation of the effects of shift work on parasomnia prevalence. *Chronobiol Int* 2021;38:1500-6.
33. Chellappa SL, Morris CJ, Scheer FA. Circadian misalignment increases mood vulnerability in simulated shift work. *Sci Rep* 2020;10:18614.
34. Xu M, Yin X, Gong Y. Lifestyle factors in the association of shift work and depression and anxiety. *JAMA Netw Open* 2023;6:e2328798.
35. Waterhouse J, Reilly T, Edwards B. The stress of travel. *J Sports Sci* 2004;22:946-65.
36. Weingarten JA, Collop NA. Air travel: Effects of sleep deprivation and jet lag. *Chest* 2013;144:1394-401.
37. Katz G, Knobler HY, Laibeli Z, Strauss Z, Durst R. Time zone change and major psychiatric morbidity: The results of a 6-year study in Jerusalem. *Compr Psychiatry* 2002;43:37-40.
38. McNeely E, Gale S, Tager I, Kincl L, Bradley J, Coull B, *et al.* The self-reported health of U.S. flight attendants compared to the general population. *Environ Health* 2014;13:13.
39. Leatherwood WE, Dragoo JL. Effect of airline travel on performance: A review of the literature. *Br J Sports Med* 2013;47:561-7.
40. Janse van Rensburg DC, Jansen van Rensburg A, Fowler P, Fullagar H, Stevens D, Halson S, *et al.* How to manage travel fatigue and jet lag in athletes? A systematic review of interventions. *Br J Sports Med* 2020;54:960-8.
41. Wittmann M, Dinich J, Merrow M, Roenneberg T. Social jetlag: Misalignment of biological and social time. *Chronobiol Int* 2006;23:497-509.
42. Mathew GM, Li X, Hale L, Chang AM. Sleep duration and social jetlag are independently associated with anxious symptoms in adolescents. *Chronobiol Int* 2019;36:461-9.
43. Takaesu Y, Shimura A, Komada Y, Futenma K, Ishii M, Sugiura K, *et al.* Association of sleep duration on weekdays or free days and social jetlag with job stress. *Psychiatry Clin Neurosci* 2021;75:244-9.
44. Min J, Jang TW, Lee HE, Cho SS, Kang MY. Social jetlag and risk of depression: Results from the Korea National Health and Nutrition Examination Survey. *J Affect Disord* 2023;323:562-9.
45. Smith LT, Walsh RF, Goel N, Alloy LB. Social jetlag and trajectories of mood symptoms and reward responsiveness in individuals at low-risk, high-risk, and with bipolar spectrum disorders: An ecological momentary assessment study. *Psychiatry Res* 2023;329:115499.
46. Okajima I, Komada Y, Ito W, Inoue Y. Sleep debt and social jetlag associated with sleepiness, mood, and work performance among workers in Japan. *Int J Environ Res Public Health* 2021;18:2908.
47. Friborg O, Rosevinge JH, Wynn R, Gradisar M. Sleep timing, chronotype, mood, and behaviour at an Arctic latitude (69°N). *Sleep Med* 2014;15:798-807.
48. Blume C, Garbaza C, Spitschan M. Effects of light on human circadian rhythms, sleep and mood. *Somnologie (Berl)* 2019;23:147-56.
49. Rishi MA, Ahmed O, Barrantes Perez JH, Berneking M, Dombrowsky J, Flynn-Evans EE, *et al.* Daylight saving time: An American Academy of Sleep Medicine position statement. *J Clin Sleep Med* 2020;16:1781-4.
50. Harrison Y. The impact of daylight saving time on sleep and related behaviours. *Sleep Med Rev* 2013;17:285-92.
51. Johnson KG, Marlow BA. Daylight saving time: Neurological and neuropsychological implications. *Curr Sleep Med Rep* 2022;8:86-96.
52. Borisenkov M, Tserne TA, Panev AS, Kuznetsova ES, Petrova NB, Timonin V, *et al.* Seven-year survey of sleep timing in Russian children and adolescents: Chronic 1-h forward transition of social clock is associated with increased social jetlag and winter pattern of mood seasonality. *Biol Rhythm Res* 2016;18:3-12.
53. Wescott DL, Soehner AM, Roehcklein KA. Sleep in seasonal affective disorder. *Curr Opin Psychol* 2020;34:7-11.
54. Anderson JL, Rosen LN, Mendelson WB, Jacobsen FM, Skwerer RG, Joseph-Vanderpool JR, *et al.* Sleep in fall/winter seasonal affective disorder: Effects of light and changing seasons. *J Psychosom Res* 1994;38:323-37.
55. Wirz-Justice A, Kräuchi K, Brunner DP, Graw P, Haug HJ, Leonhardt G, *et al.* Circadian rhythms and sleep regulation in seasonal affective disorder. *Acta Neuropsychiatr* 1995;7:41-3.
56. Bertrand L, d'Ortho MP, Reynaud E, Lejoyeux M, Bourgin P, Geoffroy PA. Polysomnography in seasonal affective disorder: A systematic review and meta-analysis. *J Affect Disord* 2021;292:405-15.
57. Sleep Foundation. How Much Sleep Do You Need? 2024. Available from: <https://www.sleepfoundation.org/how-sleep-works/how-much-sleep-do-we-really-need#:~:text=Healthy%20adults%20need%20at%20least%20seven%20hours%20of%20sleep%20per,need%20is%20a%20first%20step.> [Last accessed on 2024 Jul 09].
58. Meney I, Waterhouse J, Atkinson G, Reilly T, Davenne D. The effect of one night's sleep deprivation on temperature, mood, and physical performance in subjects with different amounts of habitual physical activity. *Chronobiol Int* 1998;15:349-63.
59. Caliendo R, Streng AA, van Kerkhof LW, van der Horst GT, Chaves I. Social jetlag and related risks for human health: A timely review. *Nutrients* 2021;13:4543.
60. Lee S. Naturally occurring consecutive sleep loss and day-to-day trajectories of affective and physical well-being. *Ann Behav Med* 2022;56:393-404.
61. Waters F, Chiu V, Atkinson A, Blom JD. Severe sleep deprivation causes hallucinations and a gradual progression toward psychosis with increasing time awake. *Front Psychiatry* 2018;9:303.
62. Berger M, Riemann D. Symposium: Normal and abnormal REM

- sleep regulation: REM sleep in depression-an overview. *J Sleep Res* 1993;2:211-23.
63. Becker NB, Jesus SN, João KA, Viseu JN, Martins RI. Depression and sleep quality in older adults: A meta-analysis. *Psychol Health Med* 2017;22:889-95.
 64. Zhu G, Cassidy S, Hiden H, Woodman S, Trenell M, Gunn DA, *et al.* Exploration of sleep as a specific risk factor for poor metabolic and mental health: A UK biobank study of 84,404 participants. *Nat Sci Sleep* 2021;13:1903-12.
 65. Hamilton OS, Steptoe A, Ajnakina O. Polygenic predisposition, sleep duration, and depression: Evidence from a prospective population-based cohort. *Transl Psychiatry* 2023;13:323.
 66. Li XL, Wei J, Zhang X, Meng Z, Zhu W. Relationship between night-sleep duration and risk for depression among middle-aged and older people: A dose-response meta-analysis. *Front Physiol* 2023;14:1085091.
 67. Dong L, Xie Y, Zou X. Association between sleep duration and depression in US adults: A cross-sectional study. *J Affect Disord* 2022;296:183-8.
 68. Newman AB, Spiekerman CF, Enright P, Lefkowitz D, Manolio T, Reynolds CF, *et al.* Daytime sleepiness predicts mortality and cardiovascular disease in older adults. The cardiovascular health study research group. *J Am Geriatr Soc* 2000;48:115-23.
 69. Gu S, Li Y, Jiang Y, Huang JH, Wang F. Glymphatic dysfunction induced oxidative stress and neuro-inflammation in major depression disorders. *Antioxidants (Basel)* 2022;11:2296.
 70. Rasmussen MK, Mestre H, Nedergaard M. The glymphatic pathway in neurological disorders. *Lancet Neurol* 2018;17:1016-24.
 71. Fultz NE, Bonmassar G, Setsompop K, Stickgold RA, Rosen BR, Polimeni JR, *et al.* Coupled electrophysiological, hemodynamic, and cerebrospinal fluid oscillations in human sleep. *Science* 2019;366:628-31.
 72. Clarke L, Chisholm K, Cappuccio FP, Tang NK, Miller MA, Elahi F, *et al.* Sleep disturbances and the at risk mental state: A systematic review and meta-analysis. *Schizophr Res* 2021;227:81-91.
 73. Brand S, Gerber M, Kalak N, Kirov R, Lemola S, Clough PJ, *et al.* Adolescents with greater mental toughness show higher sleep efficiency, more deep sleep and fewer awakenings after sleep onset. *J Adolesc Health* 2014;54:109-13.
 74. Short MA, Booth SA, Omar O, Ostlundh L, Arora T. The relationship between sleep duration and mood in adolescents: A systematic review and meta-analysis. *Sleep Med Rev* 2020;52:101311.
 75. Booth SA, Carskadon MA, Young R, Short MA. Sleep duration and mood in adolescents: An experimental study. *Sleep* 2021;44:zsaa253.
 76. Tomaso CC, Johnson AB, Nelson TD. The effect of sleep deprivation and restriction on mood, emotion, and emotion regulation: Three meta-analyses in one. *Sleep* 2021;44:zsaa289.
 77. Griggs S, Horvat Davey C, Howard Q, Pignatiello G, Duwadi D. Socioeconomic deprivation, sleep duration, and mental health during the first year of the COVID-19 pandemic. *Int J Environ Res Public Health* 2022;19:14367.
 78. Ben Simon E, Walker MP. Sleep loss causes social withdrawal and loneliness. *Nat Commun* 2018;9:3146.
 79. Kent RG, Uchino BN, Cribbet MR, Bowen K, Smith TW. Social relationships and sleep quality. *Ann Behav Med* 2015;49:912-7.
 80. Mushtaq R, Shoib S, Shah T, Mushtaq S. Relationship between loneliness, psychiatric disorders and physical health? A review on the psychological aspects of loneliness. *J Clin Diagn Res* 2014;8:E01-4.
 81. Holding BC, Sundelin T, Schiller H, Åkerstedt T, Kecklund G, Axelsson J. Sleepiness, sleep duration, and human social activity: An investigation into bidirectionality using longitudinal time-use data. *Proc Natl Acad Sci U S A* 2020;117:21209-17.
 82. Demichelis OP, Grainger SA, Burr L, Henry JD. Emotion regulation mediates the effects of sleep on stress and aggression. *J Sleep Res* 2023;32:e13787.
 83. Palmer CA, Alfano CA. Sleep and emotion regulation: An organizing, integrative review. *Sleep Med Rev* 2017;31:6-16.
 84. Yoo SS, Gujar N, Hu P, Jolesz FA, Walker MP. The human emotional brain without sleep – A prefrontal amygdala disconnect. *Curr Biol* 2007;17:R877-8.
 85. Goldstein AN, Walker MP. The role of sleep in emotional brain function. *Annu Rev Clin Psychol* 2014;10:679-708.
 86. Bonfils KA, Longenecker JM, Soreca I, Hammer LA, Tighe CA, Haas GL, *et al.* Sleep disorders in veterans with serious mental illnesses: Prevalence in veterans affairs health record data. *J Clin Sleep Med* 2023;19:1651-60.
 87. Palagini L, Rosenlicht N. Sleep, dreaming, and mental health: A review of historical and neurobiological perspectives. *Sleep Med Rev* 2011;15:179-86.
 88. Anderson KN, Bradley AJ. Sleep disturbance in mental health problems and neurodegenerative disease. *Nat Sci Sleep* 2013;5:61-75.
 89. Baglioni C, Battagliese G, Feige B, Spiegelhalter K, Nissen C, Voderholzer U, *et al.* Insomnia as a predictor of depression: A meta-analytic evaluation of longitudinal epidemiological studies. *J Affect Disord* 2011;135:10-9.
 90. Sofi F, Cesari F, Casini A, Macchi C, Abbate R, Gensini GF. Insomnia and risk of cardiovascular disease: A meta-analysis. *Eur J Prev Cardiol* 2014;21:57-64.
 91. Zheng B, Yu C, Lv J, Guo Y, Bian Z, Zhou M, *et al.* Insomnia symptoms and risk of cardiovascular diseases among 0.5 million adults: A 10-year cohort. *Neurology* 2019;93:e2110-20.
 92. Wu TT, Zou YL, Xu KD, Jiang XR, Zhou MM, Zhang SB, *et al.* Insomnia and multiple health outcomes: Umbrella review of meta-analyses of prospective cohort studies. *Public Health* 2023;215:66-74.
 93. Uccella S, Cordani R, Salfi F, Gorgoni M, Scarpelli S, Gemignani A, *et al.* Sleep deprivation and insomnia in adolescence: Implications for mental health. *Brain Sci* 2023;13:569.
 94. Bernert RA, Joiner TE. Sleep disturbances and suicide risk: A review of the literature. *Neuropsychiatr Dis Treat* 2007;3:735-43.
 95. Pigeon WR, Pinquart M, Conner K. Meta-analysis of sleep disturbance and suicidal thoughts and behaviors. *J Clin Psychiatry* 2012;73:e1160-7.
 96. Harris LM, Huang X, Linthicum KP, Bryen CP, Ribeiro JD. Sleep disturbances as risk factors for suicidal thoughts and behaviours: A meta-analysis of longitudinal studies. *Sci Rep* 2020;10:13888.
 97. Liu RT, Steele SJ, Hamilton JL, Do QB, Furbish K, Burke TA, *et al.* Sleep and suicide: A systematic review and meta-analysis of longitudinal studies. *Clin Psychol Rev* 2020;81:101895.
 98. McCall WV, Blocker JN, D'Agostino R Jr., Kimball J, Boggs N, Lasater B, *et al.* Insomnia severity is an indicator of suicidal ideation during a depression clinical trial. *Sleep Med* 2010;11:822-7.
 99. Rumble ME, McCall WV, Dickson DA, Krystal AD, Rosenquist PB, Benca RM. An exploratory analysis of the association of circadian rhythm dysregulation and insomnia with suicidal ideation over the course of treatment in individuals with depression, insomnia, and suicidal ideation. *J Clin Sleep Med* 2020;16:1311-9.
 100. Miller BJ, Parker CB, Rapaport MH, Buckley PF, McCall WV. Insomnia and suicidal ideation in nonaffective psychosis. *Sleep* 2019;42:zsy215.
 101. Sjöström N, Waern M, Hetta J. Nightmares and sleep disturbances in relation to suicidality in suicide attempters. *Sleep* 2007;30:91-5.
 102. Dong M, Lu L, Sha S, Zhang L, Zhang Q, Ungvari GS, *et al.* Sleep disturbances and the risk of incident suicidality: A systematic review and meta-analysis of cohort studies. *Psychosom Med* 2021;83:739-45.
 103. Liu JW, Tu YK, Lai YF, Lee HC, Tsai PS, Chen TJ, *et al.* Associations between sleep disturbances and suicidal ideation, plans, and attempts in adolescents: A systematic review and meta-analysis. *Sleep* 2019;42:zs054.
 104. Gupta MA, Simpson FC. Obstructive sleep apnea and psychiatric disorders: A systematic review. *J Clin Sleep Med* 2015;11:165-75.
 105. Kaufmann CN, Susukida R, Depp CA. Sleep apnea, psychopathology, and mental health care. *Sleep Health* 2017;3:244-9.
 106. Garbarino S, Magnavita N. Obstructive sleep apnea syndrome (OSAS), metabolic syndrome and mental health in small enterprise workers. Feasibility of an action for health. *PLoS One* 2014;9:e97188.
 107. Pauleto P, Réus JC, Bolan M, Massignan C, Flores-Mir C, Maia I, *et al.* Association between obstructive sleep apnea and health-related quality of life in untreated adults: A systematic review. *Sleep Breath* 2021;25:1773-89.
 108. Lee MR, Jung SM. Obstructive sleep apnea related to mental health, health-related quality of life and multimorbidity: A nationwide survey of a representative sample in republic of Korea. *PLoS One* 2023;18:e0287182.
 109. Bishop TM, Ashrafioun L, Pigeon WR. The association between sleep apnea and suicidal thought and behavior: An analysis of national survey data. *J Clin Psychiatry* 2018;79:17m11480.

110. Udholm N, Fuglsang M, Lundbye-Christensen S, Bille J, Udholm S. Obstructive sleep apnea and risk of suicide and self-harm: A Danish nationwide cohort study. *Sleep* 2022;45:zsab286.
111. Chu CS, Huang KL, Bai YM, Su TP, Tsai SJ, Chen TJ, *et al.* Risk of suicide after a diagnosis of sleep apnea: A nationwide longitudinal study. *J Psychiatr Res* 2023;161:419-25.
112. Tseng WC, Liang YC, Su MH, Chen YL, Yang HJ, Kuo PH. Sleep apnea may be associated with suicidal ideation in adolescents. *Eur Child Adolesc Psychiatry* 2019;28:635-43.
113. Reddy A, Mansuri Z, Vadukapuram R, Trivedi C. Increased suicidality and worse outcomes in MDD patients with OSA: A nationwide inpatient analysis of 11 years from 2006 to 2017. *J Acad Consult Liaison Psychiatry* 2022;63:46-52.
114. McCall WV, Benca RM, Rumble ME, Case D, Rosenquist PB, Krystal AD. Prevalence of obstructive sleep apnea in suicidal patients with major depressive disorder. *J Psychiatr Res* 2019;116:147-50.
115. Gupta MA, Jarosz P. Obstructive sleep apnea severity is directly related to suicidal ideation in posttraumatic stress disorder. *J Clin Sleep Med* 2018;14:427-35.
116. Timkova V, Nagyova I, Reijneveld SA, Tkacova R, Stewart RE, van Dijk JP, *et al.* Suicidal ideation in patients with obstructive sleep apnoea and its relationship with disease severity, sleep-related problems and social support. *J Health Psychol* 2018;25:1450-61.
117. Yang X, Yang J, Yang C, Niu L, Song F, Wang L. Continuous positive airway pressure can improve depression in patients with obstructive sleep apnoea syndrome: A meta-analysis based on randomized controlled trials. *J Int Med Res* 2020;48:1-13.
118. Ozturk B, Koundal S, Al Bizri E, Chen X, Gursky Z, Dai F, *et al.* Continuous positive airway pressure increases CSF flow and glymphatic transport. *JCI Insight* 2023;8:e170270.
119. Luyster FS. Impact of obstructive sleep apnea and its treatments on partners: A literature review. *J Clin Sleep Med* 2017;13:467-77.
120. Chu C, Wu S, Yu C, Chien-Ming C. Sleep disturbance and depressive tendency in bed partners of patients with obstructive sleep apnea. *Neurol Asia* 2021;28:367-75.
121. Carotenuto M. Obstructive Sleep Apnea - New Insights in the 21st Century. IntechOpen; 2024. Available from: <http://dx.doi.org/10.5772/intechopen.111133>. [Last accessed on 2024 Jul 10].
122. Ong JC, Crawford MR. Insomnia and obstructive sleep apnea. *Sleep Med Clin* 2013;8:389-98.
123. Ragnoli B, Pochetti P, Raie A, Malerba M. Comorbid insomnia and obstructive sleep apnea (COMISA): Current concepts of patient management. *Int J Environ Res Public Health* 2021;18:9248.
124. Sweetman A, Melaku YA, Lack L, Reynolds A, Gill TK, Adams R, *et al.* Prevalence and associations of co-morbid insomnia and sleep apnoea in an Australian population-based sample. *Sleep Med* 2021;82:9-17.
125. Lang CJ, Appleton SL, Vakulin A, McEvoy RD, Wittert GA, Martin SA, *et al.* Co-morbid OSA and insomnia increases depression prevalence and severity in men. *Respirology* 2017;22:1407-15.
126. Lee MH, Lee SA, Lee GH, Ryu HS, Chung S, Chung YS, *et al.* Gender differences in the effect of comorbid insomnia symptom on depression, anxiety, fatigue, and daytime sleepiness in patients with obstructive sleep apnea. *Sleep Breath* 2014;18:111-7.
127. Choi SJ, Joo EY, Lee YJ, Hong SB. Suicidal ideation and insomnia symptoms in subjects with obstructive sleep apnea syndrome. *Sleep Med* 2015;16:1146-50.
128. Mahmood Z, Van Patten R, Nakhla MZ, Twamley EW, Filoteo JV, Schiehser DM. REM sleep behavior disorder in Parkinson's disease: Effects on cognitive, psychiatric, and functional outcomes. *J Int Neuropsychol Soc* 2020;26:894-905.
129. Sleep Foundation (n.d.) Parasomnias. Available from: <https://www.sleepfoundation.org/parasomnias>. [Last accessed on 2024 Feb 04].
130. Denis D, French CC, Gregory AM. A systematic review of variables associated with sleep paralysis. *Sleep Med Rev* 2018;38:141-57.
131. Hrozanova M, Morrison I, Riha RL. Adult NREM parasomnias: An update. *Clocks Sleep* 2019;1:87-104.
132. Fleetham JA, Fleming JA. Parasomnias. *CMAJ* 2014;186:E273-80.
133. Wróbel-Knybel P, Flis M, Rog J, Jalal B, Wolowski L, Karakuta-Juchnowicz H. Characteristics of sleep paralysis and its association with anxiety symptoms, perceived stress, PTSD, and other variables related to lifestyle in selected high stress exposed professional. *Int J Environ Res Public Health* 2022;19:7821.
134. Denis D. Relationships between sleep paralysis and sleep quality: Current insights. *Nat Sci Sleep* 2018;10:355-67.
135. Akkaoui MA, Lejoyeux M, d'Ortho MP, Geoffroy PA. Nightmares in patients with major depressive disorder, bipolar disorder, and psychotic disorders: A systematic review. *J Clin Med* 2020;9:3990.
136. Lemyre A, Bastien C, Vallières A. Nightmares in mental disorders: A review. *Dreaming* 2019;29:144-66.
137. Hedström AK, Bellocco R, Hössjer O, Ye W, Trolle Lagerros Y, Åkerstedt T. The relationship between nightmares, depression and suicide. *Sleep Med* 2021;77:1-6.
138. Leung AK, Leung AA, Wong AH, Hon KL. Sleep terrors: An updated review. *Curr Pediatr Rev* 2020;16:176-82.
139. Thompson A, Lereya ST, Lewis G, Zammit S, Fisher HL, Wolke D. Childhood sleep disturbance and risk of psychotic experiences at 18: UK birth cohort. *Br J Psychiatry* 2015;207:23-9.
140. Laganière C, Gaudreau H, Pokhvisneva I, Kenny S, Bouvette-Turcot AA, Meaney M, *et al.* Sleep terrors in early childhood and associated emotional-behavioral problems. *J Clin Sleep Med* 2022;18:2253-60.
141. Germain A. Sleep disturbances as the hallmark of PTSD: Where are we now? *Am J Psychiatry* 2013;170:372-82.
142. Lancel M, van Marle HJ, Van Veen MM, van Schagen AM. Disturbed sleep in PTSD: Thinking beyond nightmares. *Front Psychiatry* 2021;12:767760.
143. Abenza-Abildua MJ, Suárez-Gisbert E, Lores-Gutiérrez V, Algarra-Lucas C, Gómez-Aceña A, Navacerrada-Barrero FJ, *et al.* Anxiety and depression in patients with narcolepsy. *J Sleep Res* 2022;32:e13812.
144. Ruoff CM, Reaven NL, Funk SE, McGaughey KJ, Ohayon MM, Guilleminault C, *et al.* High rates of psychiatric comorbidity in narcolepsy: Findings from the Burden of Narcolepsy Disease (BOND) study of 9,312 patients in the United States. *J Clin Psychiatry* 2017;78:171-6.
145. Tadrous R, O'Rourke D, Mockler D, Broderick J. Health-related quality of life in narcolepsy: A systematic review and meta-analysis. *J Sleep Res* 2021;30:e13383.
146. Shen Z, Shuai Y, Mou S, Shen Y, Shen X, Yang S. Case report: Cases of narcolepsy misdiagnosed as other psychiatric disorders. *Front Psychiatry* 2022;13:942839.
147. Schiappa C, Scarpelli S, D'Atri A, Gorgoni M, De Gennaro L. Narcolepsy and emotional experience: A review of the literature. *Behav Brain Funct* 2018;14:19.
148. Becker PM. The biopsychosocial effects of restless legs syndrome (RLS). *Neuropsychiatr Dis Treat* 2006;2:505-12.
149. Seeman MV. Why are women prone to restless legs syndrome? *Int J Environ Res Public Health* 2020;17:368.
150. Yilmaz O, Şengül Y, Şengül HS, Parlakkaya FB, Öztürk A. Investigation of alexithymia and levels of anxiety and depression among patients with restless legs syndrome. *Neuropsychiatr Dis Treat* 2018;14:2207-14.
151. Winkelman JW, Finn L, Young T. Prevalence and correlates of restless legs syndrome symptoms in the Wisconsin sleep cohort. *Sleep Med* 2006;7:545-52.
152. Ulfberg J, Nyström B, Carter N, Edling C. Prevalence of restless legs syndrome among men aged 18 to 64 years: An association with somatic disease and neuropsychiatric symptoms. *Mov Disord* 2001;16:1159-63.
153. Didriksen M, Allen RP, Burchell BJ, Thøner LW, Rigas AS, Di Angelantonio E, *et al.* Restless legs syndrome is associated with major comorbidities in a population of Danish blood donors. *Sleep Med* 2018;45:124-31.
154. Kallweit U, Werth E, Seiz A, Sefidan S, Dahmen N, Manconi M, *et al.* Psychiatric comorbidities in restless legs syndrome. *J Neuropsychiatry Clin Neurosci* 2016;28:239-42.
155. Zhuang S, Na M, Winkelman JW, Ba D, Liu CF, Liu G, *et al.* Association of restless legs syndrome with risk of suicide and self-harm. *JAMA Netw Open* 2019;2:e199966.
156. Chenini S, Barateau L, Guiraud L, Denis C, Rassu AL, Lopez R, *et al.* Depressive symptoms and suicidal thoughts in restless legs syndrome. *Mov Disord* 2022;37:812-25.
157. Miller MA, Howarth NE. Sleep and cardiovascular disease. *Emerg Top*

- Life Sci 2023;7:457-66.
158. Chaddha A, Robinson EA, Kline-Rogers E, Alexandris-Souphis T, Rubenfire M. Mental health and cardiovascular disease. *Am J Med* 2016;129:1145-8.
 159. De Hert M, Detraux J, Vancampfort D. The intriguing relationship between coronary heart disease and mental disorders. *Dialogues Clin Neurosci* 2018;20:31-40.
 160. Deschênes SS, Burns RJ, Graham E, Schmitz N. Depressive symptoms and sleep problems as risk factors for heart disease: A prospective community study. *Epidemiol Psychiatr Sci* 2019;29:e50.
 161. Matsuda R, Kohno T, Kohsaka S, Shiraishi Y, Katsumata Y, Hayashida K, *et al.* Psychological disturbances and their association with sleep disturbances in patients admitted for cardiovascular diseases. *PLoS One* 2021;16:e0244484.
 162. Cheng M, Lei X, Zhu C, Hou Y, Lu M, Wang X, *et al.* The association between poor sleep quality and anxiety and depression symptoms in Chinese patients with coronary heart disease. *Psychol Health Med* 2022;27:1347-56.
 163. Cai H, Wang XP, Yang GY. Sleep disorders in stroke: An update on management. *Aging Dis* 2021;12:570-85.
 164. Niu S, Liu X, Wu Q, Ma J, Wu S, Zeng L, *et al.* Sleep quality and cognitive function after stroke: The mediating roles of depression and anxiety symptoms. *Int J Environ Res Public Health* 2023;20:2410.
 165. Fulk GD, Boyne P, Hauger M, Ghosh R, Romano S, Thomas J, *et al.* The impact of sleep disorders on functional recovery and participation following stroke: A systematic review and meta-analysis. *Neurorehabil Neural Repair* 2020;34:1050-61.
 166. Joa KL, Kim WH, Choi HY, Park CH, Kim ES, Lee SJ, *et al.* The effect of sleep disturbances on the functional recovery of rehabilitation inpatients following mild and moderate stroke. *Am J Phys Med Rehabil* 2017;96:734-40.
 167. Iddagoda MT, Inderjeeth CA, Chan K, Raymond WD. Post-stroke sleep disturbances and rehabilitation outcomes: A prospective cohort study. *Intern Med J* 2020;50:208-13.
 168. Fan XW, Yang Y, Wang S, Zhang YJ, Wang AX, Liao XL, *et al.* Impact of persistent poor sleep quality on post-stroke anxiety and depression: A national prospective clinical registry study. *Nat Sci Sleep* 2022;14:1125-35.
 169. Byun E, Kohen R, Becker KJ, Kirkness CJ, Khot S, Mitchell PH. Stroke impact symptoms are associated with sleep-related impairment. *Heart Lung* 2020;49:117-22.
 170. Dong L, Brown DL, Chervin RD, Case E, Morgenstern LB, Lisabeth LD. Pre-stroke sleep duration and post-stroke depression. *Sleep Med* 2021;77:325-9.
 171. Babkair LA, Huri H, Alharbi W, Turkistani Y, Alaslani R, Alandijani N, *et al.* The association between sleep quality and depressive symptoms among stroke survivors and caregivers. *Healthcare (Basel)* 2023;12:58.
 172. Ludwig R, D'Silva L, Vaduvathiryan P, Rippee MA, Siengsucon C. Sleep disturbances in the acute stage of concussion are associated with poorer long-term recovery: A systematic review. *PMR* 2020;12:500-11.
 173. Theadom A, Cropley M, Parmar P, Barker-Collo S, Starkey N, Jones K, *et al.* Sleep difficulties one year following mild traumatic brain injury in a population-based study. *Sleep Med* 2015;16:926-32.
 174. McAllister-Deitrick J, Trbovich AM, Broglio SP, McCrear M, McAllister TW, Kontos AP. Effect of diagnosed sleep disorders on baseline concussion symptom, cognitive, and balance assessments in collegiate athletes. *Am J Sports Med* 2020;48:991-9.
 175. Waller CS, Pettibone JC, Pomerantz AM. Amplifying factors in the proposed relationship between sleep-wake dysfunction and post-concussion syndrome pathogenesis. *SN Compr Clin Med* 2020;2:526-30.
 176. Lang CP, Boucaut A, Guppy M, Johnston LM. Children with cerebral palsy: A cross-sectional study of their sleep and their caregiver's sleep quality, psychological health and well-being. *Child Care Health Dev* 2021;47:859-68.
 177. Whitney DG, Warschawsky SA, Peterson MD. Mental health disorders and physical risk factors in children with cerebral palsy: A cross-sectional study. *Dev Med Child Neurol* 2019;61:579-85.
 178. Wayne S, McCaughy E, Holley S, Annaz D, Hill CM. Sleep problems in children with cerebral palsy and their relationship with maternal sleep and depression. *Acta Paediatr* 2012;101:618-23.
 179. Sun H, Xie Z, Dong C, Fu C, Hao W, Wang Q, *et al.* Mood index, sleep duration and risk of all-cause and cause-specific dementia: A prospective analysis of 429,761 UK biobank participants. *J Public Health* 2023. [Ahead of Print]. [doi: 10.1007/s10389-023-020170w].
 180. Figueiro MG, Sahin L, Kalsher M, Plitnick B, Rea MS. Long-term, all-day exposure to circadian-effective light improves sleep, mood, and behavior in persons with dementia. *J Alzheimers Dis Rep* 2020;4:297-312.
 181. Mitolo M, Tonon C, La Morgia C, Testa C, Carelli V, Lodi R. Effects of light treatment on sleep, cognition, mood, and behavior in Alzheimer's disease: A systematic review. *Dement Geriatr Cogn Disord* 2018;46:371-84.
 182. Bromundt V, Wirz-Justice A, Boutellier M, Winter S, Haberstroh M, Terman M, *et al.* Effects of a dawn-dusk simulation on circadian rest-activity cycles, sleep, mood and well-being in dementia patients. *Exp Gerontol* 2019;124:110641.
 183. Figueiro MG, Plitnick B, Roohan C, Sahin L, Kalsher M, Rea MS. Effects of a tailored lighting intervention on sleep quality, rest-activity, mood, and behavior in older adults with Alzheimer disease and related dementias: A randomized clinical trial. *J Clin Sleep Med* 2019;15:1757-67.
 184. Giorelli AS, Passos P, Carnaval T, Gomes Mda M. Excessive daytime sleepiness and epilepsy: A systematic review. *Epilepsy Res Treat* 2013;2013:629469.
 185. Winsor AA, Richards C, Bissell S, Seri S, Liew A, Bagshaw AP. Sleep disruption in children and adolescents with epilepsy: A systematic review and meta-analysis. *Sleep Med Rev* 2021;57:101416.
 186. Adem K, Kassem T, Birhanu A, Abate A. Sleep quality and associated factors among peoples with epilepsy who have a follow-up at Amanuel Mental Specialized Hospital, Addis Ababa, Ethiopia, 2019: An institutional based cross-sectional study. *Psychiatry J* 2020;2020:1-9.
 187. Gutter T, Callenbach PM, Brouwer OF, de Weerd AW. Prevalence of sleep disturbances in people with epilepsy and the impact on quality of life: A survey in secondary care. *Seizure* 2019;69:298-303.
 188. Harrison L, Wilson S, Heron J, Stannard C, Munafò MR. Exploring the associations shared by mood, pain-related attention and pain outcomes related to sleep disturbance in a chronic pain sample. *Psychol Health* 2016;31:565-77.
 189. Koffel E, Krebs EE, Arbisi PA, Erbes CR, Polusny MA. The unhappy triad: Pain, sleep complaints, and internalizing symptoms. *Clin Psychol Sci* 2016;4:96-106.
 190. Dzierzewski JM, Griffin SC, Ravyts S, Rybarczyk B. Psychological interventions for late-life insomnia: Current and emerging science. *Curr Sleep Med Rep* 2018;4:268-77.
 191. Finan PH, Smith MT. The comorbidity of insomnia, chronic pain, and depression: Dopamine as a putative mechanism. *Sleep Med Rev* 2013;17:173-83.
 192. Zambelli Z, Halstead EJ, Fidalgo AR, Dimitriou D. Good sleep quality improves the relationship between pain and depression among individuals with chronic pain. *Front Psychol* 2021;12:668930.
 193. Lin D, Huang X, Sun Y, Wei C, Wu A. Perioperative sleep disorder: A review. *Front Med (Lausanne)* 2021;8:640416.
 194. Kontinen H, Sjöholm K, Jacobson P, Svensson PA, Carlsson LM, Peltonen M. Prediction of suicide and nonfatal self-harm after bariatric surgery: A risk score based on sociodemographic factors, lifestyle behavior, and mental health: A nonrandomized controlled trial. *Ann Surg* 2021;274:339-45.
 195. Tang NK, Lereya ST, Boulton H, Miller MA, Wolke D, Cappuccio FP. Nonpharmacological treatments of insomnia for long-term painful conditions: A systematic review and meta-analysis of patient-reported outcomes in randomized controlled trials. *Sleep* 2015;38:1751-64.
 196. Sharkey KM, Crawford SL, Kim S, Joffe H. Objective sleep interruption and reproductive hormone dynamics in the menstrual cycle. *Sleep Med* 2014;15:688-93.
 197. Kim T, Nam GE, Han B, Cho SJ, Kim J, Eum DH, *et al.* Associations of mental health and sleep duration with menstrual cycle

- irregularity: A population-based study. *Arch Womens Ment Health* 2018;21:619-26.
198. Shuster AE, Simon KC, Zhang J, Sattari N, Pena A, Alzueta E, *et al.* Good sleep is a mood buffer for young women during menses. *Sleep* 2023;46:zsad072.
 199. Meers JM, Nowakowski S. Sleep, premenstrual mood disorder, and women's health. *Curr Opin Psychol* 2020;34:43-9.
 200. Lin PC, Ko CH, Lin YJ, Yen JY. Insomnia, inattention and fatigue symptoms of women with premenstrual dysphoric disorder. *Int J Environ Res Public Health* 2021;18:6192.
 201. Eichling PS, Sahni J. Menopause related sleep disorders. *J Clin Sleep Med* 2005;1:291-300.
 202. Lee J, Han Y, Cho HH, Kim MR. Sleep disorders and menopause. *J Menopausal Med* 2019;25:83-7.
 203. Tandon VR, Sharma S, Mahajan A, Mahajan A, Tandon A. Menopause and sleep disorders. *J Midlife Health* 2022;13:26-33.
 204. Fabbrini M, Aric  , Tramonti F, Condurso R, Carnicelli L, De Rosa A, *et al.* Sleep disorders in menopause: Results from an Italian multicentric study. *Arch Ital Biol* 2015;153:204-13.
 205. Hou SY, Chiu CJ, Shea JL, Wang CL, Tang HH, Kuo PC, *et al.* Sleep and rest-activity rhythms for women at different menopausal statuses: The role of mental health. *Menopause* 2024;31:138-44.
 206. Neutzling AL, Leite HM, Paniz VM, de Bairoos FS, Dias da Costa JS, Olinto MT. Association between common mental disorders, sleep quality, and menopausal symptoms: A population-based study in Southern Brazil. *Menopause* 2020;27:463-72.
 207. Cunningham JE, Shapiro CM. Cognitive behavioural therapy for insomnia (CBT-I) to treat depression: A systematic review. *J Psychosom Res* 2018;106:1-12.
 208. Hertenstein E, Trinca E, Wunderlin M, Schneider CL, Z  st MA, Feh  r KD, *et al.* Cognitive behavioral therapy for insomnia in patients with mental disorders and comorbid insomnia: A systematic review and meta-analysis. *Sleep Med Rev* 2022;62:101597.

APPENDIX 1

Search Terms

- Sleep
- Sleep Disorder
- Sleep Quality
- Sleep Quantity
- Sleep Length
- Sleep Architecture
- Narcolepsy
- Sleep Apnoea
- Obstructive Sleep Apnoea
- Restless Leg Syndrome
- Parasomnias
- COMISA
- Mental Health
- Anxiety
- Depression
- Bipolar Disorder
- Paranoid Schizophrenia
- Cardiovascular Disease
- Stroke
- Dementia
- Cerebral Palsy
- Concussion
- Menstrual Cycle
- Menopause
- Pain
- Epilepsy